



Quant Week 3

2/13/25

A decorative geometric shape in the top-left corner, consisting of a white parallelogram and a dark blue parallelogram.

Agenda



1. Icebreaker Question
2. Brainteasers
3. Shorting
4. Margin
5. Options
6. Project



Ice Breaker

Favorite holiday



A decorative geometric shape in the top-left corner, consisting of a white parallelogram and a dark blue parallelogram.

Daily Sequence

-5, -4, 0, 9, 25, 50?





Daily Sequence

3, 4, 2^2 , 5, $10/2$, $60/12$, ?



Brain Teaser - Roulette



Brain Teaser - Roulette



You and a friend are playing russian roulette with a toy revolver.

A (toy) bullet is put into a (toy) revolver with 6 chambers, the bullet is equally likely to be in any hole. The barrel is randomly spun ONCE.

If you get the bullet you lose

Would you rather go first or second?





Brain Teaser - Roulette Part 2



Would you rather go first or second?

First Player: 1st, 3rd, or 5th spin.

First Player: $1/6 + 5/6 * 4/5 * 1/4 + 5/6 * 4/5 * 3/4 * 2/3 * 1/2 = 0.5$

Second Player: 2nd, 4th, or 5th spin

Second Player: $5/6 * 1/5 + 5/6 * 4/5 * 3/4 * 1/3 + 5/6 * 4/5 * 3/4 * 2/3 * 1/2 * 1 = 0.5$

Equal probabilities. Intuition: The bullet is either in slot 1,2, or 3, or 2, 4, or 6 so it's fixed from the beginning

What if we spin the barrel after every shot?



Brain Teaser - Roulette Part 2



What if we spin the barrel after every shot?

Now every shot has $1/6$ probability

The probability that you lose at all is p , so the probability your opponent loses is $1-p$

$$p = 1 \cdot 1/6 + (1-p) \cdot 5/6$$

$$p = 6/11$$

This means that you should go second, your odds of losing are $5/11$



Brain Teaser - Roulette Part 3



You now have 2 bullets, you have the option to spin the barrel if you want. Should you spin the barrel?

If you spin, the probability of losing is $\frac{2}{6}$

If you don't spin the probability of losing is $\frac{2}{5}$

So yes, spin the barrel



Brain Teaser - Roulette Part 4



There are still 2 bullets, but they're in consecutive positions now. If your opponent survives the first round should you spin the barrel?

Lets say the bullets are in positions 5 and 6

This means if your opponent survives, he either hit 1, 2, 3, or 4.

This means you will hit either 2, 3, 4, or 5, so you will have a 1 in 4 chance

If you spin, the probability of losing is still $2/6$

If you don't spin the probability of losing is $1/4$

So no, don't spin the barrel

Shorting





Betting Against an Asset



Suppose AAPL is trading at \$100, and you believe that it's true value is closer to \$150. Given that assets tend towards their true value, a way you could profit off of this belief is just buying AAPL. If your belief is accurate, you will eventually gain \$50 per share that you purchase.

Now suppose AAPL is trading at \$100, but you believe that it should be valued at \$50. There is clearly a price discrepancy between AAPL and your opinion of it's true value, so you should be able to profit off of this.

One way might be to find an asset that has negative correlation with AAPL's stock price, and buy that. Another way is to short AAPL.



What is Shorting?



A short is a trade in which several things occur

1. A trader borrows some asset (shares of stock, bonds, options, etc.)
2. The trader then sells that asset
3. After some time, the trader buys back that asset
4. The trader returns the asset to its original owner

Example: AAPL trades at \$100, you think it's worth \$50. You short a share of AAPL, and wait. The price then drops to \$50, you close out your position (buy back a share) and walk away \$50 richer



Downsides



Buying (longing) a stock and shorting are inverse actions

- Buying has limited downside
 - Most assets can only go down to \$0
- Shorting has unlimited downside
 - The price of any given asset can be arbitrarily high

Additional downsides to shorting

- Usually you must pay a premium to whomever you are borrowing from, especially for larger orders
- Owning a share of stock entitles you to a portion of the company, and (theoretically) a portion of the profits

Margin





Buying on the margin

Buying 1000 shares at \$40

You have \$20k, so you must borrow the rest.

Your margin is equity/stock value, this situation is 50% margin, or also 2x leverage

Assets	Liabilities + Equity
\$40,000 (1000*\$40)	\$20,000 Loan
	\$20,000 Equity



Buying on the margin

Price goes to \$50

Margin of equity to assets has dropped to 0.33

Usually a minimum margin exists. If it's hit, you will be forced to sell

Assets	Liabilities + Equity
\$50,000 (1000*\$50)	\$20,000 Loan
	\$30,000 Equity

Price rise of 25% leads to a 50% gain



Buying on the margin

Price drops to \$30 (bad)

Margin of equity to assets has dropped to 0.33

Usually a minimum margin exists. If it's hit, you will be forced to sell

Assets	Liabilities + Equity
\$30,000 (1000*\$30)	\$20,000 Loan
	\$10,000 Equity

Price drop of 25% leads to a 50% loss



Short Selling

Shorting 1000 shares at \$40

Assets	Liabilities + Equity
\$40,000 (unchanging, sale already happened)	\$40,000 Loan ($1000 * \40)
\$20,000 (initial deposit)	\$20,000 Equity



Short Selling

Price falls to \$30

Assets	Liabilities + Equity
\$40,000 (unchanging, sale already happened)	\$30,000 Loan ($1000 * \30)
\$20,000 (initial deposit)	\$30,000 Equity

Options





What are Options?



Options are contracts that give the owner the right to buy or sell an asset at a given price. Similar to shorting, there are several instances in which the ability to trade these contracts might be incredibly useful.

For example, suppose you own a large chunk of Walmart, and there is an earnings call coming up. If you are worried about the result of this earnings call, you might buy options that give you the right to sell Walmart at a little bit below the current price in case the stock price drops significantly.

In this case, options are essentially acting as insurance.



Types of Options



The two primary types of options are call options (“calls”), and put options (“puts”).

Calls:

- Gives the holder the right to buy an asset at a given price
- Generally used when an investor has positive sentiment regarding an asset

Puts:

- Gives the holder the right to sell an asset at a given price
- Generally used when an investor has negative sentiment regarding an asset



Option Characteristics



There are several characteristics of an option that are important to note:

- Price: the price of the actual option contract
- Strike Price: the price that the trade would be executed at
- Expiry Date: the last date at which the option can be exercised
 - Note that American options can be exercised at any time **before or on** the expiry date, while European options can only be exercised **on** the expiry date

An important quality to note is that any single options contract is for 100 shares of the underlying stock, but the price and strike price are listed in per share numbers. For example, if an investor buys a call option on AAPL at a \$150 strike price, and then exercises it, they will have to pay \$15,000. Additionally, if the price of the contract itself is listed at \$5.00, the investor will have had to pay \$500 for the option.

Option Chain

Option Chain - Strike Price



CALLS							PUTS							
IMP VOLATILITY	VOLUME	OPEN INT	THETA	DELTA	BID	ASK	STRIKE	BID	ASK	DELTA	THETA	OPEN INT	VOLUME	IMP VOLATILITY
59.64%	0	6	-0.02	0.97	31.20	31.40	70	0.13	0.14	-0.02	-0.01	226	33	54.12%
53.47%	0	6	-0.02	0.96	26.25	26.50	75	0.22	0.24	-0.03	-0.01	3,258	21	49.73%
48.66%	1	72	-0.03	0.93	21.50	21.70	80	0.41	0.43	-0.06	-0.02	6,773	27	45.97%
44.42%	6	315	-0.03	0.89	16.85	17.05	85	0.79	0.81	-0.11	-0.03	9,664	77	43.10%
42.21%	8	495	-0.05	0.81	12.60	12.80	90	1.49	1.52	-0.18	-0.04	2,929	693	40.86%
39.98%	276	5,557	-0.06	0.70	8.85	9.00	95	2.72	2.76	-0.30	-0.05	5,153	56	39.28%
38.48%	633	16,235	-0.06	0.56	5.85	5.90	100	4.60	4.70	-0.44	-0.06	15,850	532	38.15%
37.54%	781	7,833	-0.06	0.41	3.50	3.60	105	7.30	7.45	-0.59	-0.06	381	88	37.59%
37.00%	3,737	4,307	-0.05	0.28	2.02	2.05	110	10.80	10.90	-0.72	-0.05	283	1	37.05%
36.74%	72	4,713	-0.04	0.17	1.09	1.10	115	14.80	15.00	-0.82	-0.04	9	0	37.38%
37.15%	13	892	-0.03	0.10	0.56	0.59	120	19.30	19.50	-0.89	-0.03	16	0	38.22%
37.64%	7	291	-0.02	0.06	0.30	0.31	125	24.00	24.25	-0.93	-0.02	11	0	39.89%
38.90%	3	179	-0.01	0.04	0.16	0.18	130	28.90	29.10	-0.95	-0.02	2	0	41.50%



In the Money vs Out of the Money



When looking at a particular set of options, we can divide them into those that are In The Money (ITM) and those that are Out of the Money (OTM). These indicate whether or not the stock has inherent value right now.

ITM calls are calls that have strikes below the current price of the asset. If we have an ITM call, say for example, a \$100 strike call on AAPL while it's trading at \$125, we can execute immediately and make profit. Similarly, ITM puts are puts that have strikes above the current price.

OTM calls have strikes above the current price of the asset, and would result in lost money if executed. Similarly, OTM puts have strikes below the current price.

Greek Values





What are the Greeks?

The Greek values are metrics that are used to determine the performance of options in relation to their underlying asset.

These are the first partial derivatives of various factors influencing the asset value.

The primary greek values are:

Delta Δ , Theta Θ , Gamma Γ , Vega v , and Rho ρ

Note: Beta is not included here.



Greek Values

Delta Δ

Theta Θ

Gamma Γ

Vega v

Rho ρ

Note: **Beta** is not included here.



Beta β (not a Greek)

$$\text{Beta coefficient}(\beta) = \frac{\text{Covariance}(R_e, R_m)}{\text{Variance}(R_m)}$$

Even though Beta is a greek letter, it is not considered a greek.

It doesn't represent a partial derivative of the underlying asset like the other values do.

Beta represents the volatility of an asset compared to the rest of the market.

In terms of statistics, this is very similar to variance but is normalized to the market as a whole

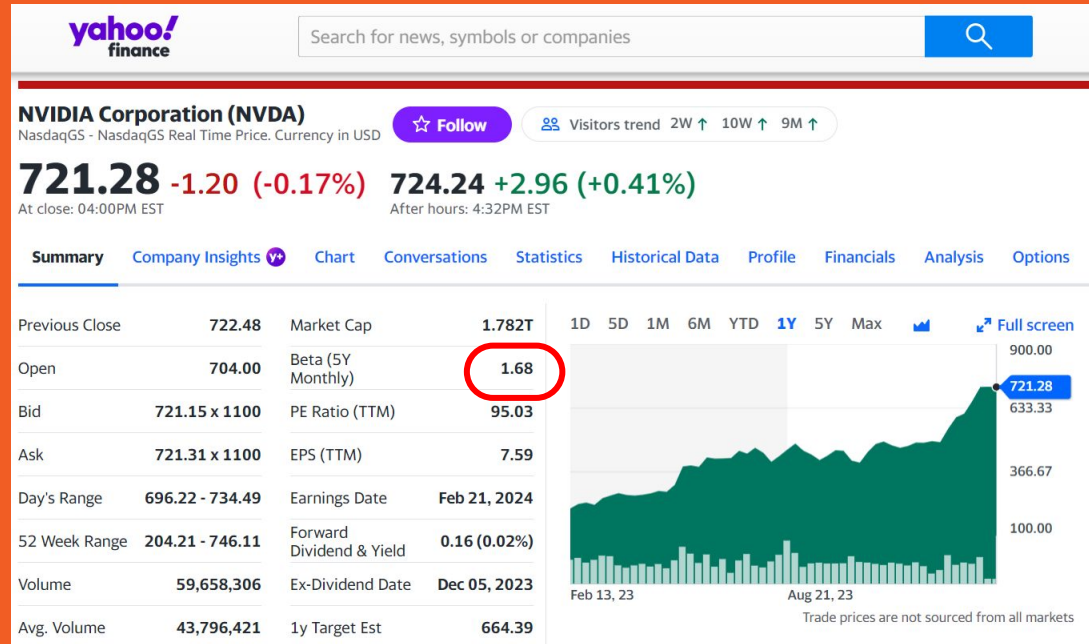
A beta of greater than 1 means the asset is more volatile than the market

A beta of less than 1 means the asset is less volatile than the market

Look at NVDA's Beta

Beta is 1.68

Means that NVDA is more volatile than the market, about 1.68x more volatile





Delta Δ

Estimates the change in price of the derivative/option based on a \$1 move in the price of the underlying asset

This value is typically not above 1.0

Example,

A stock price increases by \$2.

A call option with a strike price of \$80 that is currently trading at \$20 has a delta of 0.8
What will the new price of this be?

\$21.60



Option Chain - Strike Price

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38.90%	3	179	-0.01	0.04	0.16	0.18	130	28.90	29.10	-0.95	-0.02	2	0	41.50%



Theta Θ

Estimates the rate of change between price and time

Value will get larger as the option gets closer to date of expiration

In-the-money options will have a positive theta value, out of the money options will be negative

Example,

A theta value is -0.5

This means that for every day that passes, the price of the option will go down by \$0.50

Last 3 values next week...





Gamma Γ



Vega v



Rho ρ



Other Lesser Known Greeks

Lambda, Epsilon, Vomma, Vera, Zomma, and Ultima

- These are second and third derivatives of the other greeks we have talked about
- These trends can also be seen by those with a keen eye by looking at the main Greeks, but are explicitly calculated and named to be these variables
- Becoming increasingly more important in large algorithms because they are relatively easy to compute and can show hidden patterns

Project Discussion/Work Time

The background is a solid orange color. On the right side, there is a series of parallel diagonal lines running from the top right towards the bottom left. These lines are composed of alternating white and dark blue segments, creating a striped effect. Overlaid on these lines is a grid of squares, some of which are white and others are dark blue, creating a complex geometric pattern.